

FOR STUDENT SERVICES, ETHICS BOARDS AND RESEARCHERS

Evidence dossier

What published research says about the ideas behind CalmCampus, a student wellbeing app: where the evidence is strong, where it is thin, and what our KU Leuven pilot needs to test.

Version 1.0, October 2026

Sources 35 studies and reports, each checked against its publisher or repository record

CalmCampus is a wellness tool. It is not a medical device, and no study has yet tested CalmCampus itself.

SUMMARY

The short version

CalmCampus is an iPhone app that helps students notice early signs that their wellbeing is slipping and take one small step. It combines a one-minute daily check-in, sleep and activity signals compared with each student's own baseline, and short breathing, grounding and CBT/ACT-based exercises. All data stays on the phone.

WHAT WE CLAIM

- Mental health problems are common among students and often start early.
- Many students wait, or do not seek help at all.
- Changes in sleep and activity are worth noticing early.
- Short daily check-ins and brief exercises can support wellbeing.
- Keeping data on the phone answers a real privacy problem.

WHAT THE EVIDENCE SUPPORTS

- 35% of 13,984 first-years in 8 countries, including Belgium, screened positive for a lifetime common mental health condition [2]; about one in three KU Leuven freshmen reported a problem in the past year [1].
- Only 24.6% of first-years would definitely seek help [8].
- Insomnia predicts later depression and anxiety [11, 12], and improving sleep improves mental health [14].
- Digital self-help has small effects in students [21, 26].
- Most mental health apps share data with third parties [27].

WHAT IS STILL UNPROVEN

- That CalmCampus improves any outcome. No study has tested it.
- That phone signals can reliably spot early signs in an individual student. Accuracy has dropped sharply in larger, more varied samples [18].
- That students keep using it. Real-world 30-day retention of mental health apps is low [31].
- That on-device storage increases trust.
- How to measure student burnout well: no questionnaire meets current standards [7].

Numbers in square brackets refer to the reference list. Effect sizes are reported as the authors report them: by convention, g or d around 0.2 is small, 0.5 medium and 0.8 large.

Student mental health problems are common

Large international surveys, including samples from KU Leuven, find that a substantial share of students screen positive for common mental health conditions, and that these problems often start before university. Burnout complaints are widespread too, although burnout itself is hard to measure well. The WMH-ICS studies use screening scales for what they call common mental disorders; we use their term only when reporting their figures.

What the studies found

SOURCE	DESIGN	WHAT THE AUTHORS FOUND
Bruffaerts et al. (2018) [1]	Web survey, WMH-ICS; n = 4,921 first-year students, KU Leuven (Belgium)	About one in three KU Leuven freshmen reported a 12-month mental health problem, and these problems were linked to a 2.9–4.7% lower academic year percentage (0.2–0.3 GPA points).
Auerbach et al. (2018) [2]	Web surveys in 19 colleges, 8 countries incl. Belgium; n = 13,984 full-time first-year students; 45.5% pooled response rate	35% screened positive for at least one lifetime and 31% for at least one 12-month common mental disorder.
Auerbach et al. (2016) [3]	WHO WMH household surveys, 21 countries; n = 1,572 college students vs 4,178 non-students aged 18–22 (CIDI interviews)	20.3% of college students had a 12-month disorder, 83.1% of these began before entering college, and only 16.4% received any professional care for it in the past year.
Mason et al. (2025) [4]	Online screening surveys 2017–2023; n = 72,288 first-year students, 77 universities, 18 countries; weighted response rate 20.8%	65.2% screened positive for a lifetime and 57.4% for a 12-month mental disorder (DSM-5 screening scales).
Nuijen et al. (RIVM/Trimbos) (2025) [5]	National online survey, spring 2025; n = 27,446 students at 23 Dutch institutions (12 hbo, 11 wo); 5.9% of the 466,313 enrolled students took part	Five of ten mental health measures improved since 2021, yet only 23% of students were flourishing, and 53% of students with psychological problems who did not seek support within their institution said they preferred to solve problems themselves.
Abraham et al. (2024) [6]	Systematic review and meta-analysis; 44 studies, 26,500 university students, pandemic period	Pooled prevalence was 56.3% for high emotional exhaustion, 55.3% for high cynicism and 41.8% for low personal accomplishment.

SOURCE	DESIGN	WHAT THE AUTHORS FOUND
Hara et al. (2026) [7]	COSMIN systematic review; 44 studies, 33 self-report student burnout measures	No student burnout questionnaire met COSMIN criteria for recommendation: 10 were rated 'recommend against' and 23 'no conclusion'.

HOW THIS APPLIES TO CALMCAMPUS

The need is real and local: about one in three KU Leuven freshmen reported a mental health problem in the past year, with measurable effects on grades [1]. Because most conditions start before university [3], support that starts in the first weeks is reasonable. CalmCampus talks about early signs of burnout and wellbeing, not about any specific condition.

LIMITS

- The prevalence figures come from screening questionnaires, not clinical interviews, and response rates were low (45.5% pooled in 2018 [2], 20.8% in 2025 [4], 5.9% in the Dutch monitor [5]). The 2025 authors say their figures may be overestimates.
- The burnout figures are global and come from the pandemic period [6]. We found no verified peer-reviewed estimate for Belgian or Dutch students specifically.
- No student burnout questionnaire currently meets COSMIN standards [7]. Any burnout score we show students should be read as a rough indication, not a measurement.

CLAIM AREA B

Many students do not seek help, or seek it late

Most students who struggle do not get professional support, and the main reasons are attitudinal: wanting to handle things alone, preferring friends or family, and embarrassment or stigma.

What the studies found

SOURCE	DESIGN	WHAT THE AUTHORS FOUND
Ebert et al. (2019) [8]	WMH-ICS web surveys; n = 13,984 first-year students, 8 countries incl. Belgium	Only 24.6% said they would definitely seek help for a future emotional problem; the most common reasons not to were wanting to handle it alone (56.4%) and preferring to talk to friends or family (48.0%).
Gulliver et al. (2010) [9]	Systematic review; 22 studies (15 qualitative, 7 quantitative) of adolescents and young adults	Young people named stigma and embarrassment, difficulty recognising symptoms, and a preference for self-reliance as the main barriers to seeking help.

SOURCE	DESIGN	WHAT THE AUTHORS FOUND
Clement et al. (2015) [10]	Systematic review; 144 studies, 90,189 participants	Stigma had a small-to-moderate negative association with help-seeking (median $d = -0.27$) and ranked fourth among barriers, with youth among the groups most deterred.

HOW THIS APPLIES TO CALMCAMPUS

A private, self-paced tool fits the most common reason students give for not seeking help: they want to handle it themselves [8, 9]. The WMH-ICS authors themselves suggested testing whether private, self-help style internet tools are more acceptable to hesitant students [8]. In the Dutch monitor, 53% of students with psychological problems who did not use support within their institution said they preferred to solve problems themselves [5]. CalmCampus can point students to student services, but it should not become a reason to delay contact when things get worse.

LIMITS

- These studies measure intention and reported barriers, not actual help-seeking behaviour.
- A self-help app could reinforce the preference to cope alone instead of lowering the threshold to professional help. The pilot needs to measure this (see the research plan).
- The stigma review is not student-specific, and its searches end in 2011 [10].

CLAIM AREA C

Sleep and activity are linked to later mental health

Poor sleep and low physical activity tend to come before later depression and anxiety, and improving sleep improves mental health. This is why CalmCampus pays attention to sleep and activity.

What the studies found

SOURCE	DESIGN	WHAT THE AUTHORS FOUND
Lindsay et al. (2022) [11]	Longitudinal surveys at entry and end of first year; two cohorts at Queen's University, Canada ($n = 1,519$ and $n = 860$)	Insomnia symptoms at university entry predicted depressive ($\beta = 0.15$) and anxiety ($\beta = 0.16$) symptoms at year end, after controlling for baseline symptoms.
Hertenstein et al. (2019) [12]	Meta-analysis of 13 longitudinal studies, follow-up ≥ 12 months	Insomnia at baseline predicted later onset of depression (OR 2.83, 10 studies) and anxiety (OR 3.23, 6 studies).
Schuch et al. (2018) [13]	Meta-analysis of 49 prospective cohort studies, $N = 266,939$	People with high physical activity had lower odds of developing depression than those with low activity (adjusted OR 0.83; youths 0.90).

SOURCE	DESIGN	WHAT THE AUTHORS FOUND
Scott et al. (2021) [14]	Meta-analysis of 65 RCTs (72 interventions), N = 8,608	Interventions that improved sleep had a medium effect on composite mental health ($g = -0.53$), with a dose-response relationship.
Freeman et al. (2017) [15]	RCT at 26 UK universities; 3,755 students with insomnia, digital CBT for insomnia vs usual care	Digital CBT-I reduced insomnia strongly ($d = 1.11$) and also reduced depression ($d = 0.48$) and anxiety ($d = 0.33$) at 10 weeks.

HOW THIS APPLIES TO CALMCAMPUS

The strongest link is for sleep. Insomnia was associated with roughly three times the odds of later depression and anxiety [12], and in first-year students it predicted end-of-year symptoms even after accounting for symptoms at entry [11]. Trials show that improving sleep improves mental health [14], including in a large trial among UK students [15]. This supports paying attention to sleep and activity changes, compared with the student's own usual pattern.

LIMITS

- These studies used questionnaires or clinical sleep programmes, not phone-estimated sleep. Phones estimate sleep and activity imperfectly.
- The physical activity evidence is observational, so it shows association, not cause [13].
- Prediction at group level does not mean the signal is accurate for one person. A bad week of sleep during exams is common and usually not a warning sign.

CLAIM AREA D

Phone sensing and daily check-ins: promising, with clear accuracy limits

Smartphones can passively record behaviour such as sleep, movement and conversation, and short repeated questionnaires (ecological momentary assessment, EMA) can capture mood in daily life. Small studies show meaningful links with wellbeing. Larger studies show that these links often do not hold up across different people.

What the studies found

SOURCE	DESIGN	WHAT THE AUTHORS FOUND
Wang et al. (2014) [16]	Observational; 48 undergraduates, one 10-week term, continuous phone sensing plus about 8 EMA prompts per day	Phone-sensed sleep duration ($r = -0.36$) and daytime conversation frequency ($r = -0.40$) correlated with depressive symptoms (PHQ-9) at the start of term.
Rohani et al. (2018) [17]	Systematic review; 46 studies, 85 behavioural features from 17 sensor types	Time at home, sleep duration and vigorous activity correlated consistently with mood across studies, while other features (texting, screen use) pointed in contradictory directions.

SOURCE	DESIGN	WHAT THE AUTHORS FOUND
Müller et al. (2021) [18]	Secondary analysis; student sample N = 57 vs US-wide sample N = 5,262	A GPS-based depression model that looked accurate in a small student sample (AUC 0.82) was only slightly better than chance in a large mixed sample (AUC 0.57).
Adler et al. (2022) [19]	Secondary analysis combining StudentLife (students) and CrossCheck (schizophrenia) sensing data	Models trained on combined datasets were more often predictive than single-study models, but personalisation helped only one outcome and boosting sensitivity cost specificity.
Wrzus & Neubauer (2023) [20]	Meta-analysis; 477 articles, 496 samples, N = 677,536	EMA studies averaged six prompts a day over seven days with 79% compliance; the number of prompts did not predict compliance, while financial incentives raised it.

HOW THIS APPLIES TO CALMCAMPUS

StudentLife showed that phone-sensed sleep and conversation relate to students' wellbeing [16], and reviews find consistent links for time at home, sleep and activity [17]. But models that look accurate in small student samples can fall to near chance in larger, more varied groups [18]. That is one reason CalmCampus compares each student with their own baseline instead of a population model, and asks the student directly through the daily check-in. This design choice is consistent with the evidence. It has not been tested. Research EMA studies reach about 79% compliance over roughly a week, often with payment [20]. A one-minute daily check-in over a semester is a different situation, and we do not yet know how compliance will hold up.

LIMITS

- We do not claim that CalmCampus can identify burnout or any condition from phone data. Its signals are prompts for the student to reflect, not assessments.
- Personal baselines need enough days of data and can be thrown off by holidays, exams and illness. False alarms and missed changes are both expected.
- Most sensing studies are small, short and from the United States. StudentLife had 48 students in one class [16].

CLAIM AREA E

Brief digital self-help: small but real effects

Internet and app-based programmes, especially those based on CBT, have small positive effects on depression, anxiety and stress, including in university students. Breathing exercises show small-to-medium effects on self-reported stress.

What the studies found

SOURCE	DESIGN	WHAT THE AUTHORS FOUND
Harrer et al. (2019) [21]	Meta-analysis of 48 RCTs in university students	Effects were small for depression ($g = 0.18$) and stress ($g = 0.20$); anxiety ($g = 0.27$) lost significance after adjusting for publication bias, and well-being was not significant ($g = 0.15$).
Linardon et al. (2024) [22]	Meta-analysis of 176 RCTs	Apps had small but significant effects on depression ($g = 0.28$, NNT 11.5) and generalised anxiety ($g = 0.26$, NNT 12.4), larger when they included CBT elements or, for anxiety, mood monitoring.
Lattie et al. (2019) [23]	Systematic review; 89 studies in college students	47% of programmes were effective and 34% partially effective, but half of the studies reported no usability or acceptability data and only 4% examined campus-wide implementation.
Weisel et al. (2019) [24]	Meta-analysis; 19 RCTs, 3,681 adults with elevated symptoms	Standalone apps improved depression ($g = 0.33$) but showed no significant pooled effect on anxiety, and the authors concluded standalone apps cannot yet be recommended.
Fincham et al. (2023) [25]	Meta-analysis; 12 RCTs (785 adults) for stress, 20 for anxiety, 18 for depression	Breathwork was associated with lower self-reported stress than controls ($g = -0.35$, small-to-medium).
Jabs et al. (2026) [26]	Multilevel meta-analysis; 122 studies of app-based interventions in university students	Apps had small effects on anxiety ($d = 0.22$), depression ($d = 0.25$) and stress ($d = 0.28$) and a moderate effect on well-being ($d = 0.47$), with 17% attrition and 70% adherence.

HOW THIS APPLIES TO CALMCAMPUS

The exercises in CalmCampus (breathing, grounding, CBT/ACT-based) belong to families of techniques with some evidence behind them [22, 25]. In students specifically, effects are small: around $g = 0.18$ to 0.25 for depression and 0.20 to 0.28 for stress [21, 26]. The honest claim is that brief exercises can support wellbeing a little for some students, not that they solve problems.

LIMITS

- Effects are small, heterogeneous and partly due to publication bias. In the student meta-analysis, the anxiety effect was no longer significant after adjustment, and the effect on well-being was not significant [21].
- Fully standalone apps have the weakest evidence. One review concluded they cannot yet be recommended as standalone interventions [24].
- Many student studies have moderate-to-severe risk of bias, and few looked at campus-wide rollout [23].
- CalmCampus's exercises are very short. Most trial programmes were longer and more structured.

Privacy is a real problem in mental health apps

Audits repeatedly find that mental health apps send personal data to third parties, often without clear disclosure. People are least willing to share data that can identify them, such as location, and want control over deleting it.

What the studies found

SOURCE	DESIGN	WHAT THE AUTHORS FOUND
Huckvale et al. (2019) [27]	Cross-sectional technical audit of 36 top-ranked depression and smoking-cessation apps (US, Australia)	33 of 36 apps (92%) transmitted data to third parties and 29 (81%) sent data to Google or Facebook, often without disclosing it.
Iwaya et al. (2023) [28]	Static and dynamic privacy analysis of 27 top-ranked Android mental health apps	The apps showed unnecessary permissions, weak cryptography, leaks of personal data in logs and web requests, and third-party sharing.
Melcher & Torous (2020) [29]	Review of apps recommended on 60 US college counselling-centre websites (218 unique apps)	Of the apps still available, 39% had no privacy policy, 49% of those with a policy shared data with third parties, and only 16% had published efficacy studies.
Roque & Felt (2025) [30]	Online survey; N = 1,489 US adults (Prolific)	People were least willing to share data with high re-identification potential such as GPS, and 71% wanted the ability to delete all their data.

HOW THIS APPLIES TO CALMCAMPUS

CalmCampus keeps all data on the phone, with no cloud AI and no analytics. That directly avoids the third-party sharing found in most audited apps [27, 28], including apps that US counselling centres recommend to students [29]. It also fits people's stated wish to control and delete their data [30]. On-device processing is a design choice, not a research finding. These sources describe the problem; none of them shows that on-device storage increases trust or use.

LIMITS

- We found no study testing whether students trust on-device mental health apps more, or use them more.
- On-device storage has trade-offs. Data can be lost with the phone, and the university will not see aggregate data, which limits evaluation unless students opt in during a study.
- The privacy audits are snapshots from 2018 to 2022, mostly of Android apps.

Risks and criticism

Most people stop using mental health apps within weeks. Self-monitoring can occasionally increase stress or worsen mood, and most studies do not monitor harms well. A wellbeing app can also give false reassurance if a student reads a calm week of data as proof that all is fine.

What the studies found

SOURCE	DESIGN	WHAT THE AUTHORS FOUND
Baumel et al. (2019) [31]	Real-world usage panel data for 93 Android mental health apps ($\geq$ 10,000 installs)	Median retention was 3.9% at day 15 and 3.3% at day 30, and breathing-exercise apps had a median 30-day retention of 0%.
Liu et al. (2026) [32]	Meta-analysis of 79 RCTs (92 app conditions)	Even in trials, adherence was 61.8% and attrition 18.6% at post-test and 28.4% at follow-up; attrition was lower with reminders and human contact.
Borghouts et al. (2021) [33]	Systematic review of 208 articles	Common barriers to engagement were severe mental health problems, technical issues and lack of personalisation; facilitators included insight into one's health and a sense of control.
Astill Wright et al. (2025) [34]	Systematic review and meta-analysis of 77 mood-monitoring studies	Only 19% of studies reported adverse events; pooled prevalence was 4%, including increased burden or stress (4%) and mood worsening (2%).
Munir et al. (2026) [35]	Scoping review of 37 studies of self-guided digital interventions for young people (2019–2024)	Only 12 of 37 studies (32.4%) described how they assessed adverse events, leaving a gap in safety monitoring of self-help tools for young people.

HOW THIS APPLIES TO CALMCAMPUS

Real-world retention is the biggest threat to CalmCampus's value: a median 3.3% of users still used mental health apps after 30 days, and breathing apps did worst [31]. Reminders and human contact are associated with lower dropout in trials [32], which supports linking CalmCampus to student services rather than presenting it as a standalone answer. Mood monitoring produced adverse effects such as increased burden or worse mood in a small share of users, but these were badly under-reported [34], and only a third of youth self-help studies said how they measured harm [35]. CalmCampus should measure harms in its own pilot rather than assume there are none.

LIMITS

- Daily check-ins could increase rumination or worry for some students. The 4% pooled adverse-event figure comes from clinical populations and is uncertain [34].
- False reassurance is a plausible risk that we found no direct study of. A student whose phone data looks normal may still be struggling. The app should say this clearly.
- Engagement figures from trials [32] are higher than real-world figures [31]. A pilot with motivated volunteers will likely overestimate engagement.

What we do not claim

- **CalmCampus is a wellness tool, not a diagnostic or medical device.** It does not identify, assess or address any mental health condition, and it does not replace student services, a GP or a psychologist.
- **CalmCampus has not been shown to improve outcomes.** No published study has tested CalmCampus. The studies in this dossier tested other programmes, devices and populations. They support the reasoning behind the design, not the product.
- **We do not claim the app can spot burnout from phone data.** Sleep and activity changes are shown to the student as prompts to reflect. They are not scores or warnings.
- **We do not claim on-device processing makes the app safe.** It removes one specific risk, third-party data sharing, and introduces others, such as data loss.
- **We do not claim effects larger than the research shows.** Where digital self-help works, its effects in students are small [21, 26].

The KU Leuven pilot is designed to test whether the approach is acceptable, safe and useful for students. Until it reports, these statements stand.

RESEARCH PLAN

Questions for the KU Leuven pilot

These are the questions the pilot is designed to answer. No results exist yet. Design details (sample size, comparison group, measures, ethics approval and pre-registration) will be set out in a separate protocol.

1. **Engagement.** What share of students complete the daily check-in in weeks 1, 4, 8 and 12, and who stops using the app? Does use differ by year of study, gender or faculty?
2. **Acceptability and trust.** Do students understand that their data stays on the phone? Does this affect whether they enable sleep and activity signals?
3. **Feasibility of personal baselines.** How many days of data does a stable baseline need, and what share of students have enough sensor data to build one?
4. **Agreement between signals.** Do changes from a student's own sleep and activity baseline coincide with changes in their self-reported mood, energy and stress?
5. **Wellbeing.** Does wellbeing, measured with validated questionnaires, change over the semester among users compared with a comparison group?
6. **Help-seeking.** Does using the app change students' intention to seek help, or the timing of contact with student services, in either direction?
7. **Safety.** How often do students report that daily check-ins increase stress or worry, or that the app reassured them when they were not fine? How are these events recorded and followed up?
8. **Exercises.** Which exercises are used, completed and rated as helpful, and does use relate to short-term changes in self-reported stress?

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#	SOURCE	AREA	DESIGN AND SAMPLE	ACCESS
1	Bruffaerts et al. 2018	A	Web survey, WMH-ICS; n = 4,921 first-year students, KU Leuven (Belgium)	Free
2	Auerbach et al. 2018	A	Web surveys in 19 colleges, 8 countries incl. Belgium; n = 13,984 full-time first-year students; 45.5% pooled response rate	Free
3	Auerbach et al. 2016	A	WHO WMH household surveys, 21 countries; n = 1,572 college students vs 4,178 non-students aged 18–22 (CIDI interviews)	Free
4	Mason et al. 2025	A	Online screening surveys 2017–2023; n = 72,288 first-year students, 77 universities, 18 countries; weighted response rate 20.8%	Free
5	Nuijen et al. (RIVM/ Trimbos) 2025	A	National online survey, spring 2025; n = 27,446 students at 23 Dutch institutions (12 hbo, 11 wo); 5.9% of the 466,313 enrolled students took part	Free
6	Abraham et al. 2024	A	Systematic review and meta-analysis; 44 studies, 26,500 university students, pandemic period	Free
7	Hara et al. 2026	A	COSMIN systematic review; 44 studies, 33 self-report student burnout measures	Free
8	Ebert et al. 2019	B	WMH-ICS web surveys; n = 13,984 first-year students, 8 countries incl. Belgium	Free
9	Gulliver et al. 2010	B	Systematic review; 22 studies (15 qualitative, 7 quantitative) of adolescents and young adults	Free
10	Clement et al. 2015	B	Systematic review; 144 studies, 90,189 participants	Paywalled
11	Lindsay et al. 2022	C	Longitudinal surveys at entry and end of first year; two cohorts at Queen's University, Canada (n = 1,519 and n = 860)	Free
12	Hertenstein et al. 2019	C	Meta-analysis of 13 longitudinal studies, follow-up ≥ 12 months	Paywalled
13	Schuch et al. 2018	C	Meta-analysis of 49 prospective cohort studies, N = 266,939	Free
14	Scott et al. 2021	C	Meta-analysis of 65 RCTs (72 interventions), N = 8,608	Free
15	Freeman et al. 2017	C	RCT at 26 UK universities; 3,755 students with insomnia, digital CBT for insomnia vs usual care	Free
16	Wang et al. 2014	D	Observational; 48 undergraduates, one 10-week term, continuous phone sensing plus about 8 EMA prompts per day	Free
17	Rohani et al. 2018	D	Systematic review; 46 studies, 85 behavioural features from 17 sensor types	Free
18	Müller et al. 2021	D	Secondary analysis; student sample N = 57 vs US-wide sample N = 5,262	Free
19	Adler et al. 2022	D	Secondary analysis combining StudentLife (students) and CrossCheck (schizophrenia) sensing data	Free
20	Wrzus & Neubauer 2023	D	Meta-analysis; 477 articles, 496 samples, N = 677,536	Free
21	Harrer et al. 2019	E	Meta-analysis of 48 RCTs in university students	Free
22	Linardon et al. 2024	E	Meta-analysis of 176 RCTs	Free
23	Lattie et al. 2019	E	Systematic review; 89 studies in college students	Free
24	Weisel et al. 2019	E	Meta-analysis; 19 RCTs, 3,681 adults with elevated symptoms	Free
25	Fincham et al. 2023	E	Meta-analysis; 12 RCTs (785 adults) for stress, 20 for anxiety, 18 for depression	Free
26	Jabs et al. 2026	E	Multilevel meta-analysis; 122 studies of app-based interventions in university students	Free
27	Huckvale et al. 2019	F	Cross-sectional technical audit of 36 top-ranked depression and smoking-cessation apps (US, Australia)	Free

28	Iwaya et al. 2023	F	Static and dynamic privacy analysis of 27 top-ranked Android mental health apps	Free
29	Melcher & Torous 2020	F	Review of apps recommended on 60 US college counselling-centre websites (218 unique apps)	Paywalled
30	Roque & Felt 2025	F	Online survey; N = 1,489 US adults (Prolific)	Free
31	Baumel et al. 2019	G	Real-world usage panel data for 93 Android mental health apps ($\geq 10,000$ installs)	Free
32	Liu et al. 2026	G	Meta-analysis of 79 RCTs (92 app conditions)	Paywalled
33	Borghouts et al. 2021	G	Systematic review of 208 articles	Free
34	Astill Wright et al. 2025	G	Systematic review and meta-analysis of 77 mood-monitoring studies	Free
35	Munir et al. 2026	G	Scoping review of 37 studies of self-guided digital interventions for young people (2019–2024)	Free

Areas: A Prevalence and burden; B Help-seeking, stigma and barriers; C Sleep and activity as early indicators; D Passive sensing and daily check-ins; E Brief digital self-help and exercises; F Privacy, on-device processing and trust; G Risks and criticism.